

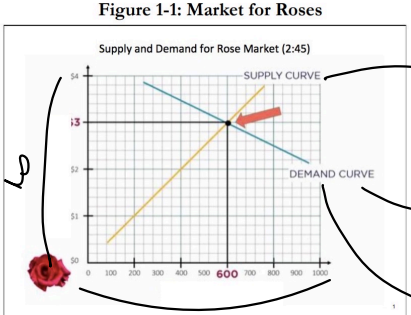
what is micro economics? the study how indiv and firms get well off in a world of scarcity.

opportunity cost → every action or inaction has a cost in terms of the next best opportunity

→ always there

Supply and demand:

Figure 1-1: Market for Roses



P

Q

Q = 200p

Q = 1800 − 400p

eq!

$Q^* = 600$

$P^* = 3$

as price goes up so does opportunity costs.

positive economics is the study of how things are

normative economics is the study of how things should be

gov makes everything (g)

capitalist → capitalism → invisible hand → inequality

socialist

communist → corruption → hard → equality

I) preferences

assumptions we are gonna make

1) completeness (that when faced with two choices are gonna have opinions)

2) transitivity

3) non-satiation (more is always better)

indifference curves (peoples preferences)

a) 2S+1C b) 1S+2C c) 2S+2C

properties of indifference curves:

1) higher better. 2) downward sloping

3) they never cross 4) completeness - 1 IC per point

Figure 2-3: Indifference curves crossing

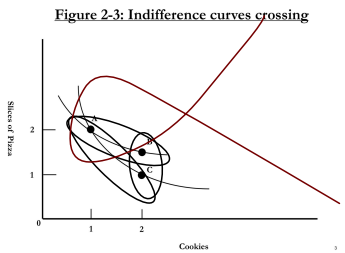


Figure 2-2: Upward-sloping indifference curve

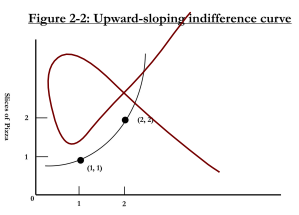
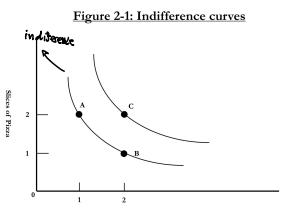


Figure 2-1: Indifference curves



utility functions

$U = \sqrt{S \cdot C}$

marginal utility

Diminishing marginal utility

Figure 2-5: Diminishing marginal utility for $U = \sqrt{P \cdot C}$

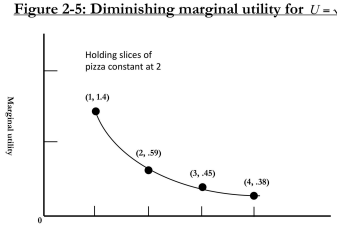


Figure 2-4: Utility levels for $U = \sqrt{P \cdot C}$

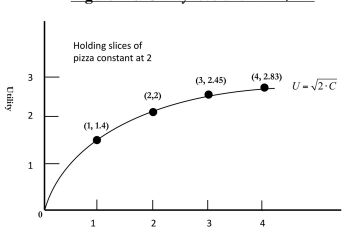
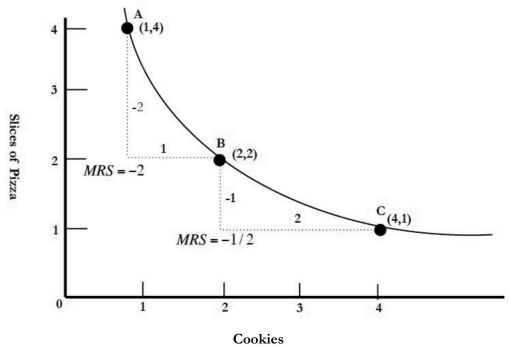


Figure 2-6: Indifference curve and MRS



$\frac{dS}{dC}$ = the good on the y-axis for the good on the x-axis = MRS

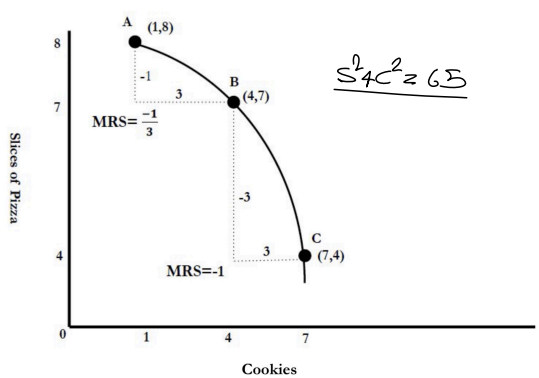
$dS \times MU_S + dC MU_C$

$\frac{dS}{dC} = MRS = -\frac{MU_C}{MU_S}$

$U = \sqrt{SC}$

$MU_C = \frac{dU}{dC} =$

Figure 2-7: Concave indifference curve



$S^2 + C^2 = 65$

example:

Starbucks ×0.90 → 5.75 ← Decreasing marginal utility

4.55 small — twice as much

budget constraints

assuming:

1. no saving 2. no borrowing 3. every penny spent

$Y = C \cdot p_c + S p_s$

$22 = .2 + .4$

$-\frac{p_c}{p_s} = -\frac{1}{2}$ (marginal rate of transformation)

p_s 4 → 8

p_c 2

Figure 3-1: Budget constraint

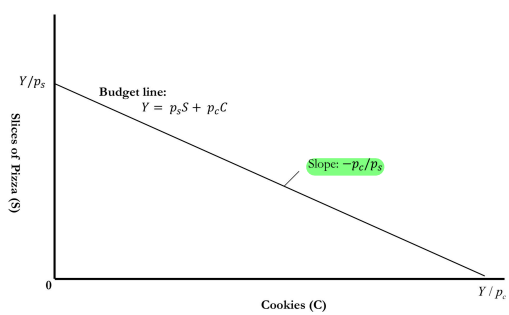


Figure 3-2: Increase in the price of pizza

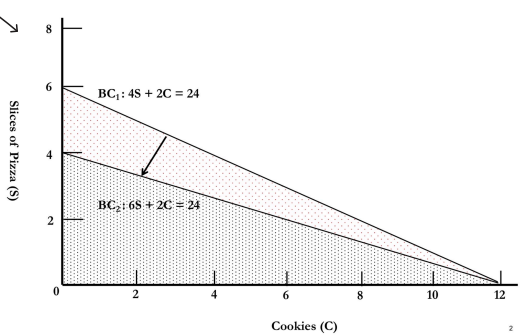


Figure 3-3: Decrease in income

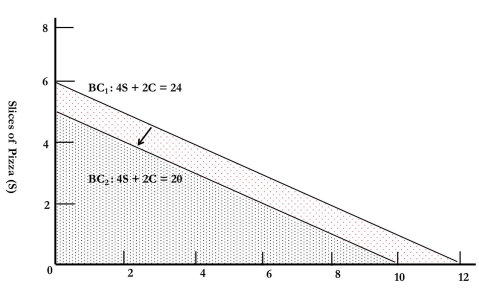
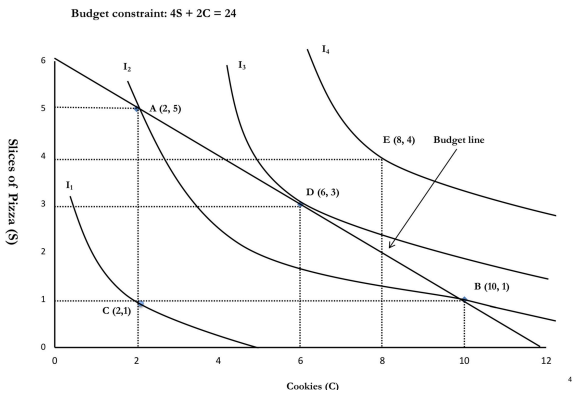


Figure 3-4: Budget constraint and indifference curves



$U = \sqrt{C \times S}$

U is the most optimal choice

$D = \sqrt{18}$ $A = \sqrt{10}$ $C = \sqrt{2}$

$E = \sqrt{32}$ $B = \sqrt{10}$

$2S + 2C = 12$

$MRS = -\frac{MU_S}{MU_S}$

$MRT = -\frac{P_C}{P_S}$

$\frac{MU_C}{MU_S} = \frac{-P_C}{P_S}$) optimal (eqⁿ of how much pleasure i get and how much i pay

$MD = MC$ $\frac{MU_C}{P_C} = \frac{MU_S}{P_S}$) bang for your buck

$S = 4$ $C = 2$ $\frac{MU_C}{MU_S} > \frac{P_C}{P_S}$ = point a because it coming from above so the MM is higher

A: $MU_S = \frac{0.5 \times S}{\sqrt{S \cdot C}}$, $MU_C = \frac{0.5 \times C}{\sqrt{S \cdot C}}$ $MRS = -2.5 > MRT = -\frac{1}{2}$

$\frac{1}{\sqrt{10}}$ $\frac{2.5}{\sqrt{10}}$

B: $MU_S = \frac{0.5 \times 1}{\sqrt{10}}$, $MU_C = \frac{0.5 \times 10}{\sqrt{10}}$ $MRS = -10 > MRT = -\frac{1}{2}$

$\frac{0.5}{\sqrt{10}}$ $\frac{5}{\sqrt{10}}$

SNAP (food stamps)

$y = 5000$ $y^* = 5500$

$p_f = p_s = 1$

x C likes shelter, doesn't mind food)

x_1 200 food, 4800 shelter

x_2 300 food, 5200 shelter

y C likes food, doesn't mind shelter)

with snap however person x can't get what they wanted.

Figure 3-5a: Cash transfer

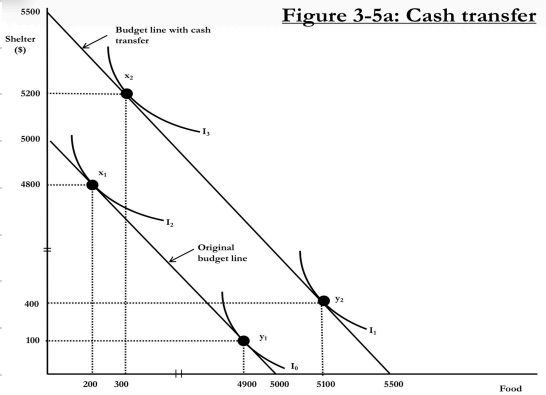
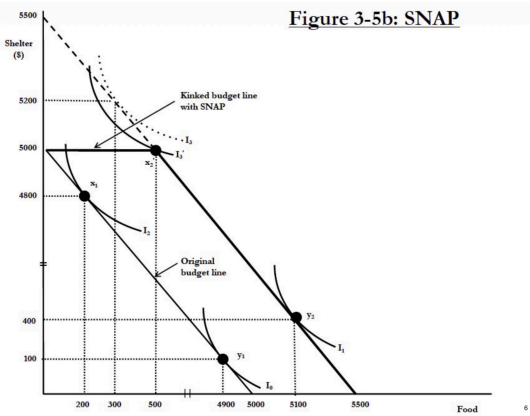


Figure 3-5b: SNAP



I) Deriving a demand curve

① Step one (solve for optimal consumption) S^*, C^*

$u = \sqrt{s \times c}$ $y = p_s \times s + p_c \times c$

$24 \quad 4 \quad 2$

$MRS = MAT$

$-\frac{s}{c} = -\frac{1}{2} = -\frac{2}{4}$

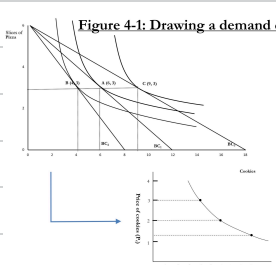
$24 = 4s + 2c$ $C^* = 6, S^* = 3$

$p_c \quad 2 \rightarrow 3$

$-\frac{s}{c} = -\frac{3}{4}$

$24 = 4s + 3c$ $C^* = 4, S^* = 3$

Figure 4-1: Drawing a demand curve

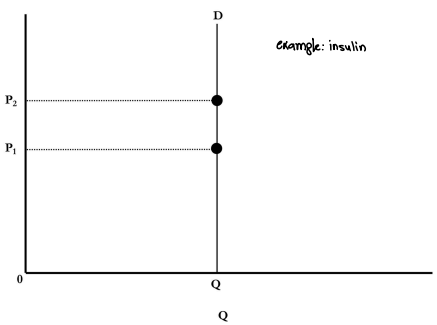


II) shape of demand curves

$$\varepsilon = \frac{\Delta Q/Q}{\Delta P/P} \approx \frac{\delta Q}{\partial P}$$

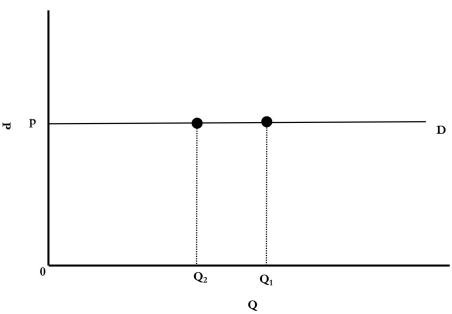
$\varepsilon = 0$

Figure 4-2: Perfectly inelastic demand



2

Figure 4-3: Perfectly elastic demand



3

III) income

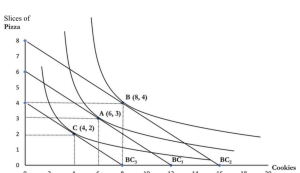
$$-\frac{S}{C} = -\frac{1}{2}$$

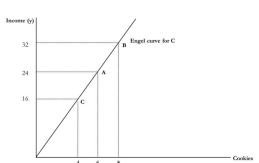
B) $82 = 4S + 8C$ $\quad C^* = 8, S^* = 4$

C) $16 = 2S + 4C$ $\quad C^* = 4, S^* = 2$

A) $24 = 3S + 6C$ $\quad C^* = 6, S^* = 3$

Figure 4-4: Generating an Engel curve





$$\gamma = \frac{\Delta Q/Q}{\Delta Y/Y} \sim \frac{dQ}{dy}$$

$\gamma > 0$ normal good $\quad \gamma > 1$ - Luxury

$\gamma < 0$ inferior $\quad \gamma < 1$ - necessity

IV) Mechanics of Δ

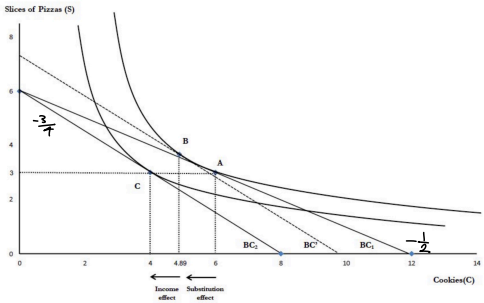
① substitution effect: ΔQ at $\bar{U}$ (has to be the opposite side of price change)

$\uparrow \frac{MUc}{MUs} = \frac{Pc}{Ps} \uparrow$

you lower the MUs and rise MUc by having fewer cookies and more slices

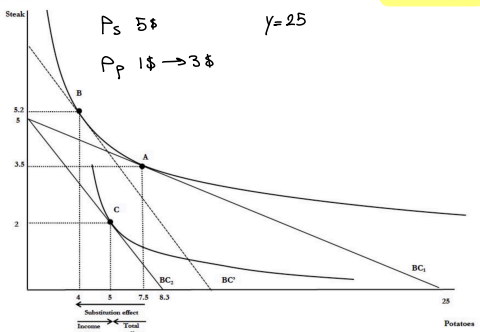
② income effect: $\Delta Q \rightarrow$ y↓ opportunity set is constricted

Figure 4-5: Income and substitution effect for a normal good



5

Figure 4-6: Income and substitution effect for an inferior good



6

Giffen good = higher price leads to higher demand (imaginary)

priceΔ	subst	Inc	Net
↑	normal ↓	↓	↓
↓	↑	↑	↑
	inferior		
↑	↓	↑	↓
↓	↑	↓	↑

I) production functions.

max $\Pi \rightarrow$ consumer Demand, cost of production, competition

minimize this cost, maximize profits

technically feasible, takes 2 things as inputs (factors of production)

production function → goods and services

Inputs:

- Labor – wage
- Capital – rental rate

both things you rent

$q = f(L, K)$ (L: Labor, K: Capital)

$Q =$ all the market

Variable (long-run) or fixed (short-run)

so lets assume L is a variable while k is fixed (short-run) but in the long run it would both be variables

II) SR production

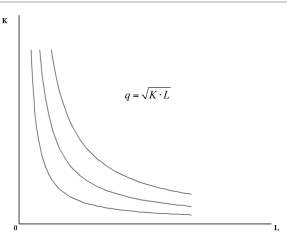
$q = f(L, \bar{K})$, $MP_L = \frac{dq}{dL}$ Lets assume MP_L is diminishing (so diminishing marginal returns)

($\bar{K}$: fixed)

III) LR production

$q = \sqrt{L \times K}$ ← isoquants

Figure 5-2a: Perfectly substitutable inputs



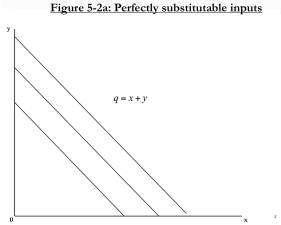
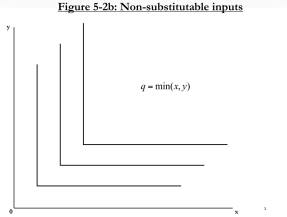


Figure 5-2b: Non-substitutable inputs



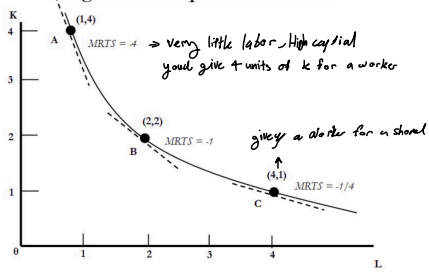
$dL \cdot MP_L + dK \cdot MP_K = 0$

$\Rightarrow \frac{dK}{dL} = MRTS = -\frac{MP_L}{MP_K}$

Let's solve MRTS for our $q = \sqrt{L \times K}$

$MP_L = \frac{0.5 \times K}{\sqrt{K \times L}}$ $MP_K = \frac{0.5 \times L}{\sqrt{L \times K}}$ $MRTS = \frac{-K}{L}$

Figure 5-3: Isoquants and MRTS



→ very little labor, high capital
you'd give 4 units of K for a worker

giving a doctor for a shovel

IV) RTS (Returns To Scale)

CRTS = $f(2L, 2k) = 2 \cdot f(L, k)$, DRTS = $f(2L, 2k) < 2 \cdot f(L, k)$, IRTS $f(2k, 2L) > 2 \cdot f(k, L)$

V) Productivity

$q = \underbrace{A}_{\text{productivity}} \cdot f(L, K)$

So where does A comes from?
1) education 2) firm research and development 3) public research

how should we spend productivity?
either goes to lower work hours or more output

who gets A? almost everybody until 1970's then all of it went to the top 1%.

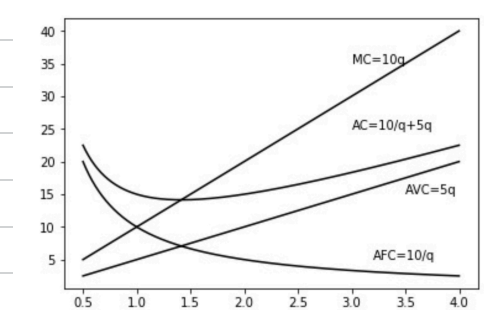
I) SR Cost curves

$q = \sqrt{L \times K}$ $C(q) = \bar{k} \cdot r + L(q) \cdot w$ example: $w = 5$ $\bar{k} = 1$ $r = 10$ $\Rightarrow L = q^2/\bar{k}$, $C = 10\bar{k} + 5q^2/\bar{k} = 10 + 5q^2 = c$ fixed costs here would be 10

Variable costs would be $5q^2$, marginal costs $= \frac{dc}{dq} = \frac{dVc}{dq}$, avg-cost $= \frac{c}{q}$

$MC = 10q$ $AC = \frac{10}{q} + 5q$ $AVC = 5q$ $AFC = 10/q$

Figure 6-1: Cost curves for $C(q) = 10 + 5q^2$



$C = \bar{k}r + Lw$

$MC = \underbrace{W}_{\text{price of labor}} \cdot \underbrace{\frac{dl}{dq}}_{\text{how many workers you need for the next unit}}$ (per unit)

$\frac{dq}{dl} = MP_L$ → how productive the next product of labor is

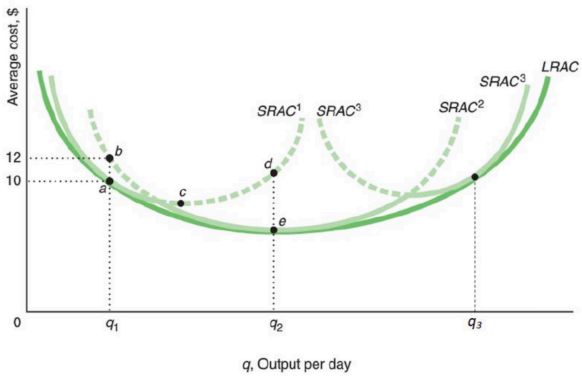
$MC = \frac{w}{mp_L}$

Sunk costs (once you make them you never get it back): they are irrelevant when decision making

II) LR/SR

III) LR costs

Figure 6-2: LRAC as envelope of SRAC



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Figure 6-3: Isocost Lines

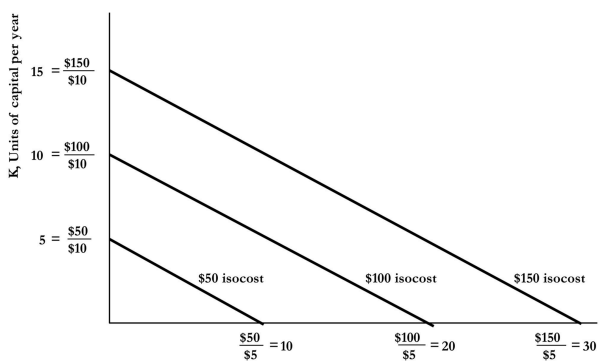
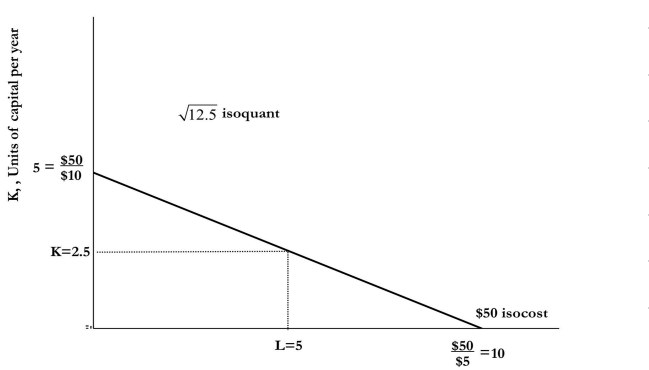


Figure 6-4: Cost Minimization



isocost line (cost line of labor and capital that help you choose within the same budget constraint)

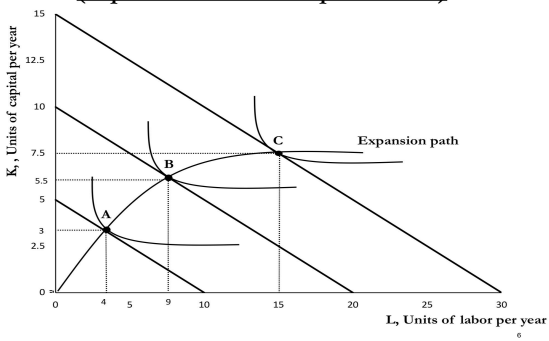
isocost = $-\frac{w}{R}$ isoquant: $\frac{-MP_L}{MP_K}$

$-\frac{K}{L} = -\frac{w}{r} = -\frac{1}{2}$

MRTS = input price ratio

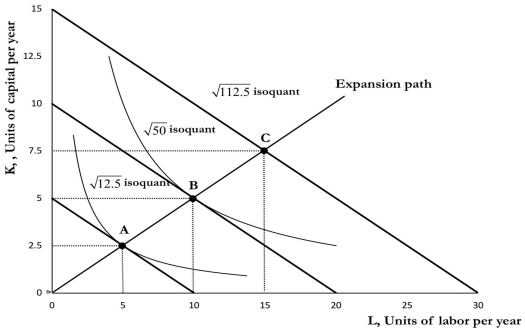
$\frac{-MP_L}{MP_K} = -\frac{w}{r}$

Figure 6-5b: Long-run expansion path (capital becomes less productive)



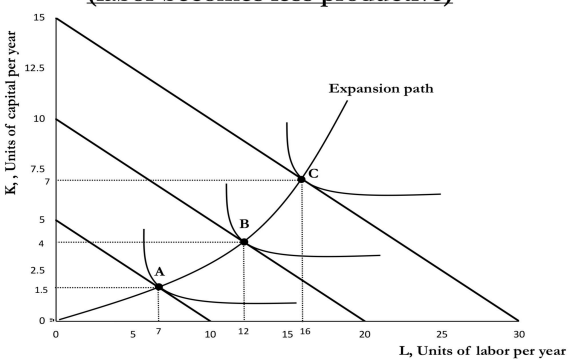
6

Figure 6-5a: Long-run expansion path (linear)



5

Figure 6-5c: Long-run expansion path (labor becomes less productive)



7

IV) measuring costs

$\pi = R - C$ (costs; revenue)

Let's say R = 60k C = 40k

60k - 40k = 20k you could've made much more

Competition

IDSR TT max, perfect competition

A) Assumptions to a perfect competition

1. all firms are price takers

1.1. identicall products 1.2. Full knowledge of prices

1.3. no transaction costs

Maximize profits:

$\pi(q) = R(q) - c(q)$

$\frac{dR}{dq} = \frac{dc}{dq}$

$\frac{\Delta P}{\Delta q} = \frac{\Delta MC}{\Delta q}$

$C = 10 + 5q^2 \quad p = 30$

$AC = \frac{10}{3} + 5.3 = 18.33$

$\pi/q = 30 - 18.33 = 11.67$

Lets say the firm pays 10$ worth of taxes on each q

$C = 5q^2 + 10q + 10$ so then $MC = 10q + 10$

So then $p = MC$ would be met at $q^* = 2$

Shutdown.

$MC = 10q = 10 \quad q^* = 1$

$\pi = 10 - (10 + 5) = -5$ if you shutdown $q = 0$ would give $\pi = -10$

so your better off just losing 5$ because then you can fix it

$pq < VC$

if $P > \frac{VC}{2}$, then you will keep producing, but here the ladder can't happen. $C = 10 + 5q^2 \rightarrow VC = 5q^2 \rightarrow AVC = 5q = P/2$ $10q = p$

Figure 7-1: Supply shift with perfectly elastic demand

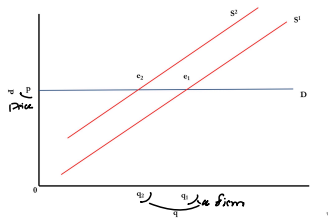
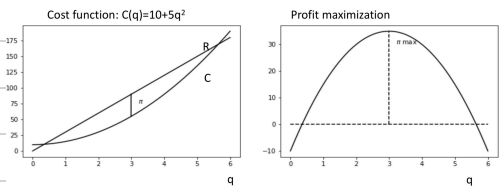


Figure 7-2: Profit maximization



MC=C'(q) = 10q, price = 30

Figure 7-3: Cost curves for $C=10+5q^2$

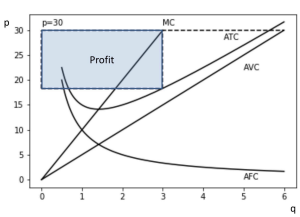


Figure 7-4: Cost curves with tax

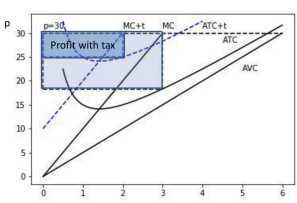


Figure 7-5: The firm's supply decision

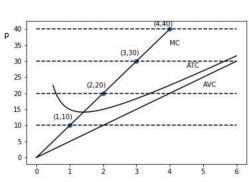


Figure 7-6: Short-run firm supply curve

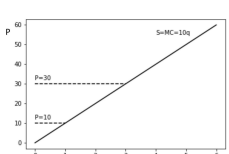
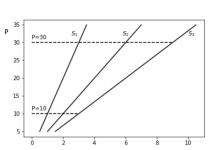


Figure 7-7: Short-run market supply curve



Decive SR eql

A) get the supply

1) start with a production function: $q = \sqrt{L \times k}$ 2) get input prices $w = 5, r = 10, \bar{k} = 1$ 3) find cost function: $C(q) = 10 + 5q^2$ 4) $MC = 10q = (q)' = P \Rightarrow \boxed{q = \frac{P}{10}}$

where $AVC < P$

$\rightarrow q = \sqrt{L \times k} \rightarrow L = \frac{q^2}{k} \rightarrow c = 10\bar{k} + \frac{5q^2}{k} \xrightarrow{\bar{k}=1} C = 10 + 5q^2$

Decive Market supply curve

$n = 6$ = number of firms $Q = 6q = \frac{3}{5}P$

Decive Demand curve

$q = 48 - P$ eql: $48 - P = \frac{3}{5}P \Rightarrow Q^d = 18 \Rightarrow q^d = 3$

$Q = \frac{3}{5}P \rightarrow P^* = 30$

Firm supply

$P = MC \quad 30 = 10q \rightarrow q^* = 3$

I) Pc in LR

- Free entry and exit

Figure 8-2: Firm exit

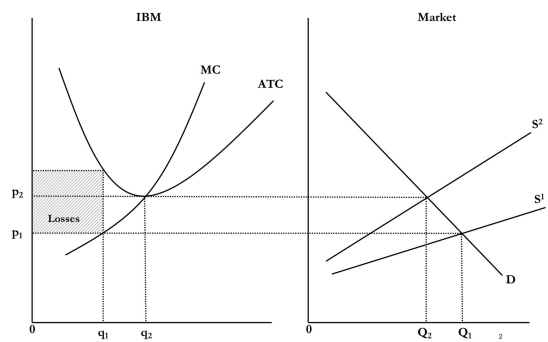
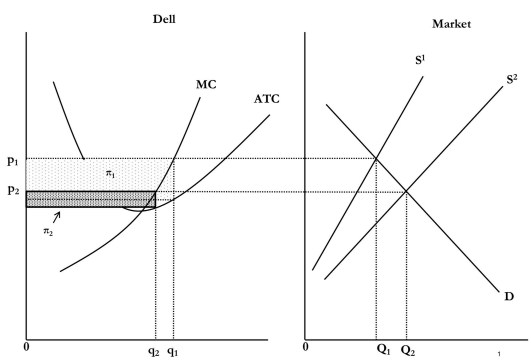
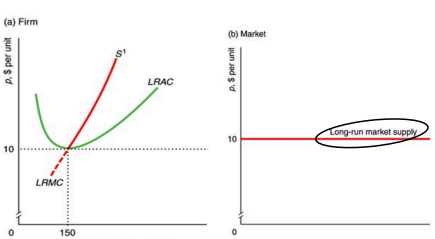


Figure 8-1: Firm entry



in the long run profits are zero

Figure 8-3: Long-run market supply with identical firms



each firm will be pushed to produce where $MR = MC = ATC$

$\pi = 0 \quad MC = ATC \quad \min ATC$

Well then how those Apple etc.. actually make money? it's because of a number of complications

$\pi > 0$ $\pi = 0$

1- limited entry. barriers to entry (sunk cost) 2- firms are identical (diff/and minimum avg cost) 3. Input prices depend on quantity

Figure 8-4: International LR market supply curve for cotton

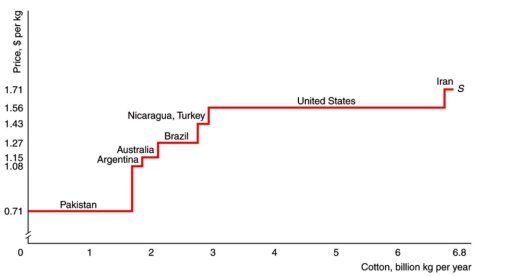


Figure 8-5: Shifts in labor market in the long-run

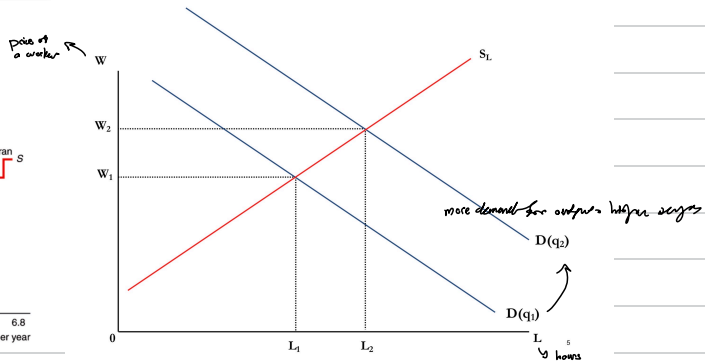
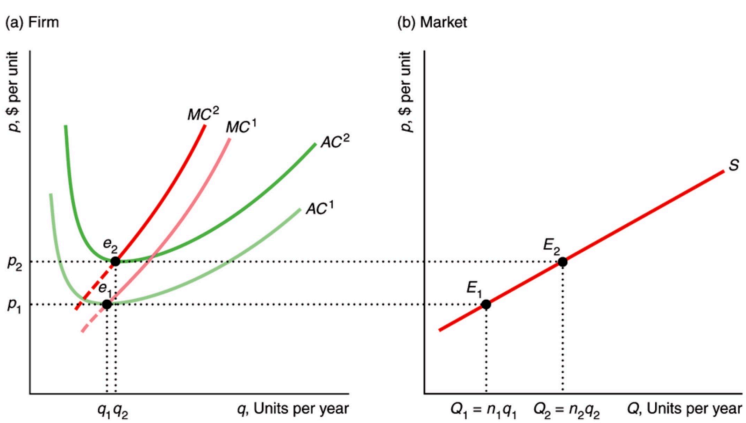
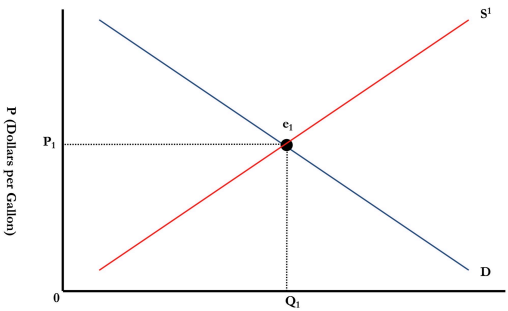


Figure 8-6: Long-run market supply in an increasing cost market



1) shocks to D+S

Figure 9-1: Market equilibrium



say big corp go viral, then D₁ would shift outwards.

Figure 9-2: Market Equilibrium with Demand shift

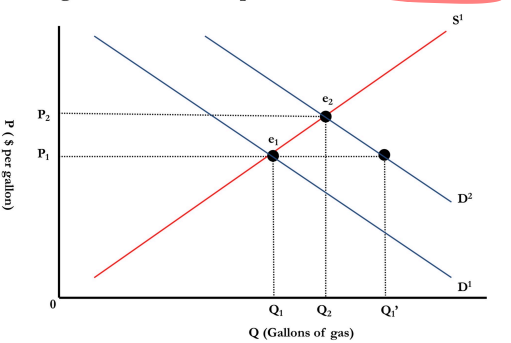
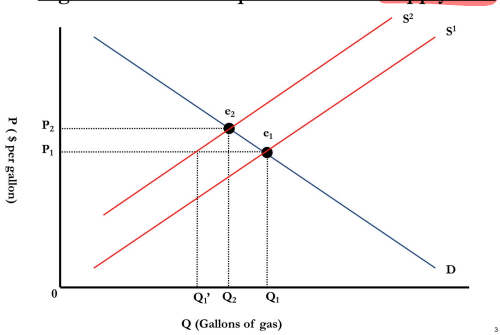


Figure 9-3: Market equilibrium with supply shift



notice how on both P goes up while Q goes up in one and lower in the other. you can determine under a P shift was from supply side or D side solely from P

D curves

- income change = Y Δ — change in a price of complement = ΔP complement — change in a price of a substitute = Δ P sub — change in prefers

1) shapes of Demand and supply

Figure 9-4: Perfectly inelastic demand

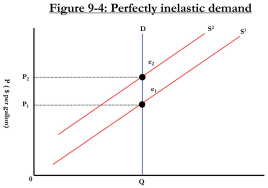
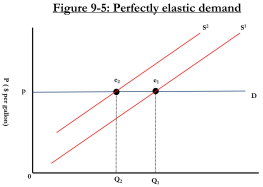


Figure 9-5: Perfectly elastic demand



III consumer and supplier surplus

welfare econ (wellbeing of society)

compensating variation → revealed preferences

Figure 9-6: Consumer surplus

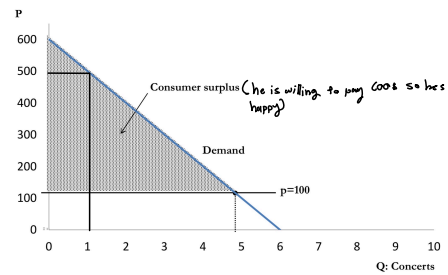


Figure 9-7: Consumer surplus in the market for Gas

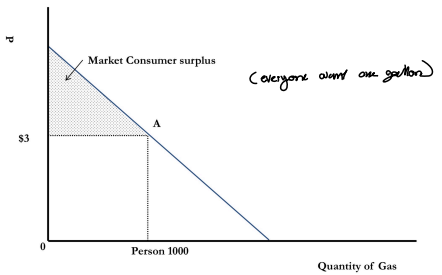


Figure 9-8: Change in consumer surplus after a price increase

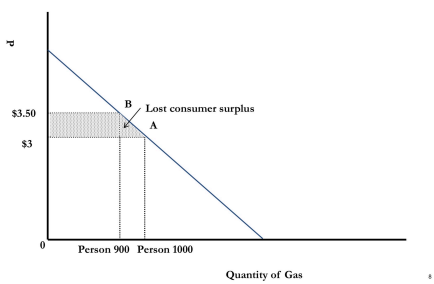


Figure 9-9: Consumer surplus and elasticity of demand

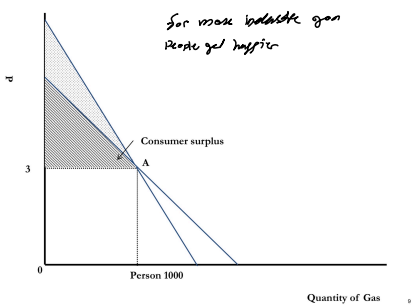


Figure 9-10: One Firm Producer surplus

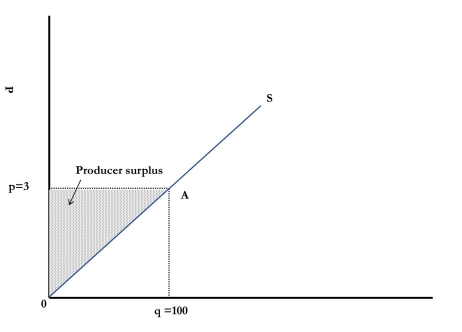
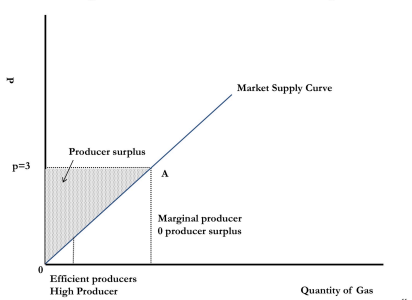


Figure 9-11: Market Producer surplus



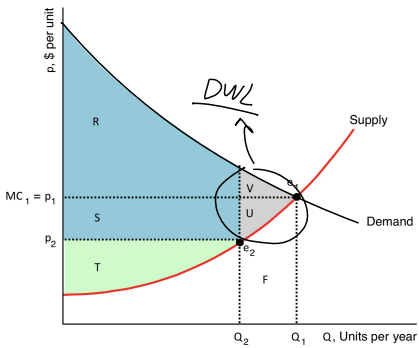
Competition maximizes allSur

Figure 9-12: Impact of a decrease in output on welfare

CS = R+V PS = S+T+U

W = CS + PS

by locking price to P_2 we transferred S to consumer but the sum is lower



12

Figure 10-2(i): Supply shift

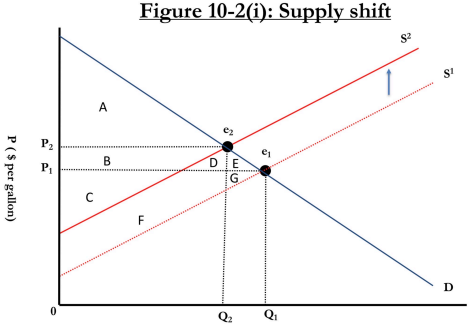
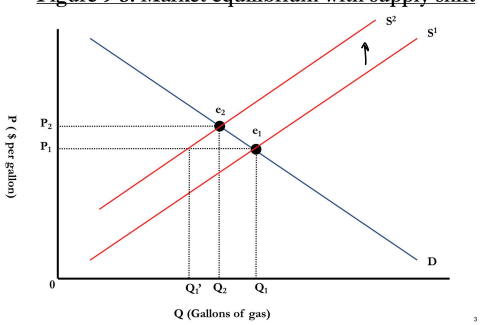


Figure 9-3: Market equilibrium with supply shift



3

before A+B+D+F = cs C+G+F = ps

after A+B = cs C+B = ps Lost = F+D+E+G

befor price-lock V+y = cs W+Z+X = ps

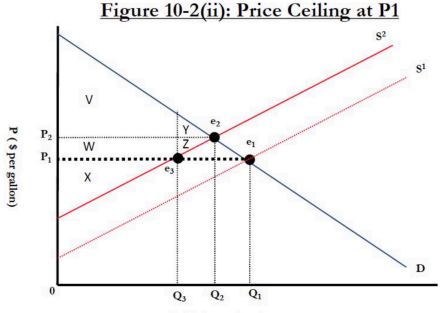
after price-lock V+W = cs X = ps Dead weight Loss

↖ efficiency loss

2- allocative inefficiency

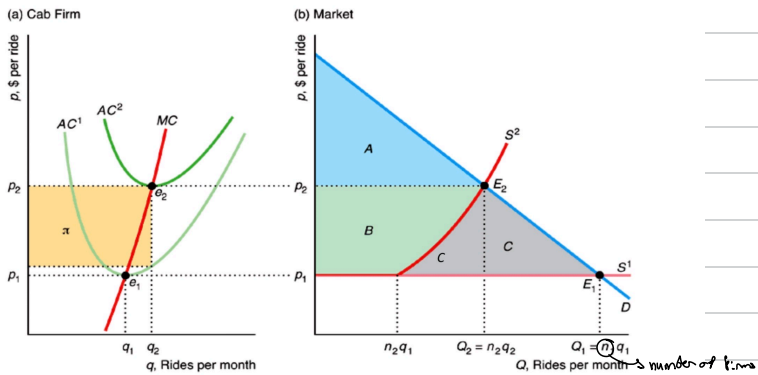
equity

Figure 10-2(ii): Price Ceiling at P1



3

Figure 10-3: Effect of a Restriction on the Number of Cabs



	No Restrictions	Restrictions	Change
Consumer Surplus, *CS*	$A + B + C$	A	$-B - C = \Delta CS$
Producer Surplus, *PS*	0	B	$B = \Delta PS$
Welfare, $W = CS + PS$	$A + B + C$	$A + B$	$-C = \Delta W = DWL$

assume identical cabs minimum average costs at q_1

occupational licencing

Government only regulate number of cab

at n_1

CS: A+B+C PS: ∅

at n_2

CS: A PS: B

1. monopoly (the exact opposite of a free and perfectly competitive market)

price makers

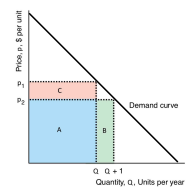
marginal revenue

Cannot price discriminate

Figure 11-1: Average and marginal revenue for a competitive firm



Figure 11-2: Average and marginal revenue for a monopolistic firm



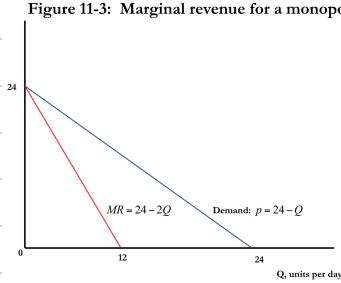
they have to lower price

poisoning effect

$p_2 - (p_1 - p_2) \cdot q_1 = MR$ (for the new unit)

Loss

Figure 11-3: Marginal revenue for a monopolist

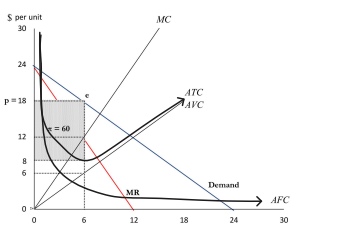


$Q = 24 - P$ $P = 24 - Q$

$P \cdot Q = 24Q - Q^2$ $MR = 24 - 2Q$

$R = P \cdot Q \Rightarrow MR = \frac{dc}{dq} + p + \frac{dp}{dq} \cdot q$) hard to lower price to sell that extra unit

Figure 11-4: Profit Maximization for a Monopolist



$MR = MC$ $\pi = R - C$ $MR - MC = 0$

$C = 12 + q^2$ $MC = 24 - 2q$ $MC = 2q$

$24 - 2q = 2q$ $q^* = 6$ $p = 18$ $p^* = 24 - q^* = 18$ $6 \cdot 18 - 48 = 60$

if they sell 7 units however they lose $58 < 60$

$q = 7$ price = 17 $7 \cdot 17 - (12 + 49) = 58$

Market power

$MR = p + \frac{dp}{dq} \cdot q$ $p + \frac{dp}{dq} \cdot \frac{q}{p} \cdot p$

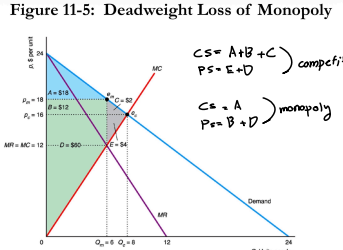
$MR = p(1 + \frac{1}{\varepsilon}) = MC$

$\frac{MC}{p} = (1 + \frac{1}{\varepsilon})$ $\frac{p - MC}{p} = \text{Markup} = -\frac{1}{\varepsilon}$

$\varepsilon \uparrow$ = replaceable $\varepsilon \downarrow$ = irreplaceable

welfare monopoly

Figure 11-5: Deadweight Loss of Monopoly

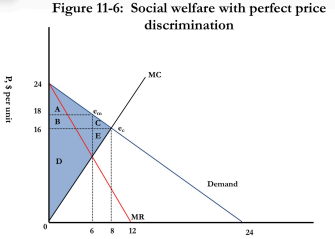


$CS = A + B + C$
$PS = E + D$
) competitive

$CS = A$
$PS = B + D$
) monopoly

→ market failure

Figure 11-6: Social welfare with perfect price discrimination



Origin of monopolies

Natural.

There are huge costs to enter so its a natural monopoly.

-delivery -property -patent (pros: gives insentive to invest, cons: its a monopoly)

-regulating monopolies

When regulations go wrong

Figure 12-1: Cost curves for a water utility



Figure 12-3: Setting regulated price too low

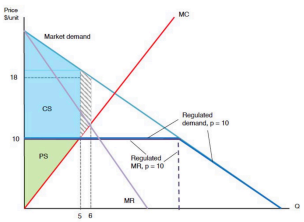


Figure 12-2: Optimal price regulation

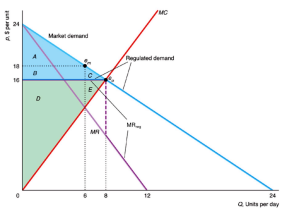
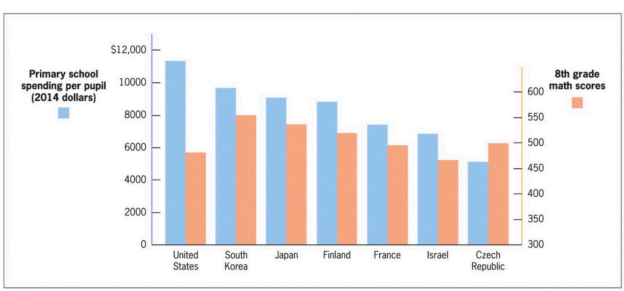


Figure 12-4: Education Spending and Outcomes Around the World



Gruber, *Public Finance and Public Policy*, 6e, © 2019 Worth Publishers

After removing regulations for airlines

prices ↓ 1/3 Routes ↑ service quality ↓↓ it excluded people who want Low quality - Low prices

Safety no Δ

oligopoly

think of automotive, you can collude together (cartel), obv there are oligopolies that aren't a cartel (more realistic) → opec* a shared monopoly

II) game theory:

n - firms playing a game

- strategy - eql (when does the game stop) → nash eql (where no player wants to change their strat given everyone's strat)

prisoners dilemma:

	prisoner B silent	prisoner B talks
prisoner A silent	A=1, B=1	A=5, B=0
prisoner A talks	A=0, B=5	A=3, B=3

Dominant strat: no matter what the other guy does, the strat they do

Dominant cooperate strat: both stay silent

Dominant strat: they both talk (so competition left them all worse)

example:

coke \ pepsi	no ads	ads
no ads	C: 8 bill, P: 8 bill	C: 0 bill, P: 11 bill
ads	C: 11 bill, P=0 bill	C: 3 bill, P: 3 bill

dominant co: dont advertise

dominant no-co: advertise

repeated game: (pepsi pov: no ad? coke no ad, yes ad? coke ad forever)

ad: 11+3+3+.....

no ad: 8+8+8+.....

a model of non-cooperative oligopoly (on model in a whole class of models)

III) Cournot model (enrich the rivals model)

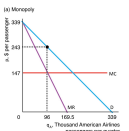
flights from my, chic 2 firms (united, american) how do they decide p or q

the cournot eql is a set of quantities for each firm such that holding other q's constant, no firm can raise π by deviating

Demand curve: $339-Q$ $MC=147$ $MR=339-Q^2$ $MR=339-2Q$

$MR=Mc$ $339-2Q=147$ $192=2Q$ $Q_m=96$, $p_m=243$

optimal q for monopoly



$MC_A=MC_u$

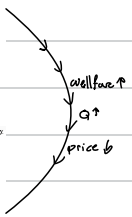


Figure 13-2: Profit-maximizing output under duopoly



lets say american knows united's $Q=64$, what should they do?

well we have to calculate residual demand

american

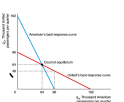
so $D=339-q_A-64 \rightarrow D=275-q_A$ $MR=275-2q_A$ $Mc=147$

now optimal q_A: $275-2q_A=147$ $q_A=64$ $p_A=211$

So what united does is ask "if united flies x what's my best response, and we get this"

its symetric so each response curve is the flip of the other

Figure 13-3: American and United's best response curves



the math:

we solve for each firms res demand (symmetric so only do one)

American

$D=339-q_A-q_u$

$T=339q_A-q_A^2-q_uq_A$

$MT=339-2q_A-q_u$

$Mc=147$

$MT=Mc$ $\therefore q_A=96-\frac{1}{2}q_u$ thus $q_u=96-\frac{1}{2}q_A$ $\therefore q_A=64$ $q_u=64$ $\therefore Q=128$ $p=211$

if it was perfect competition

$p=Mc$ $p=147$ $339-q=147$ $q^*=192$

so $q_{monopoly}:96$, $q_{oligopoly}:128$, $q_{free}:192$